

Do In-Match Hydration Breaks Alter Match Momentum?

A Within-Match Case-Crossover Analysis of the 2026 FIFA World Cup

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Abstract: The 2026 FIFA World Cup was the first to mandate hydration breaks, brief stoppages near the twenty-second minute of each half, in every match regardless of weather, prompting a widespread claim that they blunt the momentum of whichever side is dominant. We test this with a within-match case-crossover design that uses non-break minutes of the same match as self-matched controls, differencing out every match-level characteristic and adjusting only for game time, scoreline, and pre-break momentum state. A naive before-and-after comparison is misleading: momentum reverts through the stoppage as if play had continued, decaying at essentially the same rate whether or not a break occurs, so the break neither creates nor prevents the fade; it instead forgoes the interpolated minutes of pressure the dominant side would otherwise have generated, a removal that leaves no trace in net expected goals. Defining the target as a case-crossover effect on the treated, local to the observed break minutes, we find the average effect only weakly identified, resting on a time-trend extrapolation into a window with no untreated minutes, whereas break-by-covariate interactions are stable. The average effect is small and near zero, but with 99 matches the design lacks the power to reject a small effect, so we read the results by effect size and compatibility interval rather than by statistical significance. The estimates are not uniform across match states: for a comfortably-leading side of average strength the implied effect is a momentum loss (near -4 points on a pre-break level of about $+20$). We report this state-dependent signal as exploratory and in need of replication.

Keywords: hydration break; case-crossover design; match momentum; causal inference; association football.

1 Introduction

For the first time, the 2026 FIFA World Cup made in-match hydration breaks universal: a stoppage of roughly three minutes was administered near the 22nd minute of each half in every match, regardless of conditions. The policy drew immediate and pointed criticism from coaches and pundits, much of it framed explicitly in the language of momentum. Emma Hayes, the United States

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women’s head coach and a television pundit during the tournament, observed that a break “is advantageous for the team losing momentum,” adding “that’s why I call them momentum breaks” (ESPN, 2026a). The England head coach Thomas Tuchel objected that the stoppage “interrupts and changes the identity of a football match,” dividing the game “almost in four quarters” (ESPN, 2026b), while the United States men’s head coach Mauricio Pochettino said plainly that it serves to “cut the rhythm of the game” (Vertelney, 2026). The tactical claim embedded in these reactions is that a break disproportionately harms the team that has built up pressure, allowing a defending side to recover its structure and dissipating the attacking side’s momentum. This paper asks whether that claim is supported by the within-tournament evidence, and delineates the boundary of what the data can answer.

1.1 Related literature

Environmental heat stress is a first-order determinant of physical output in football. Nassis et al. (2015), using 2014 World Cup data, showed that higher wet-bulb globe temperature (WBGT) was associated with reduced distance covered, motivating in-match mitigation. The WBGT index, which integrates temperature, humidity, radiant heat, and air movement, has a long occupational history and well-documented limitations when transported into sport (Budd, 2008). Consensus guidance for competing in the heat (Racinais et al., 2015) and football-specific mitigation frameworks (Gouttebauge et al., 2023) recommend scheduled cooling or hydration breaks above high WBGT thresholds, of which the 2014 World Cup implementation is the canonical example. The physiological case for the intervention is well supported: controlled work shows that even brief in-play cooling breaks reduce thermal strain during football in the heat (Chalmers et al., 2019), and simulation studies confirm the efficacy of the specific FIFA cooling-break heat policy (Brown et al., 2025). Whether such stoppages also perturb the flow of the contest is a separate and far less studied question.

The most direct precedent is the mid-half drinks break introduced when football resumed behind closed doors during the 2020 pandemic. Analyses of that restart documented measurable shifts in match dynamics, from game actions and tempo (García-Aliaga et al., 2022) to the near-disappearance of home advantage (Almeida and Leite, 2021; Fischer and Haucap, 2021; Link and Anzer, 2022); the additional stoppage, however, was only one element of a bundled change that also removed crowds and expanded substitutions, and was not itself isolated as a treatment. More generally, the frequency and duration of stoppages in football have been quantified (Siegle and Lames, 2012), and the type of interruption is known to shape how much time is added and how the match state evolves (Li et al., 2026). The cleanest causal evidence on whether a deliberate stoppage interrupts momentum comes from other sports: Gibbs et al. (2022) estimate the effect of an NBA timeout on an opponent’s scoring run, and Den Hartigh and Gernigon (2018) show experimentally how a time-out disrupts the build-up of psychological momentum in table tennis. We nonetheless found no peer-reviewed work estimating the causal effect of a drinks or cooling break on subsequent scoring or momentum in football, and existing discussion of “momentum breaks” remains journalistic, which is the gap the present paper addresses.¹

The construct at stake is psychological momentum, the perception of progression toward a goal that in turn alters subsequent effort and performance (Vallerand et al., 1988; Gernigon et al., 2010), and its study in football is now substantial. Elite players report that they perceive momentum as real and consequential (Redwood-Brown et al., 2018), and a last-minute equalizer in knock-out

¹The operational parameters of the 2026 policy (the near-22’ timing and the approximately three-minute duration) are documented in FIFA and IFAB material rather than in the peer-reviewed literature, and are cited to those primary sources.

football has been shown to shift the subsequent course of a match (Den Hartigh et al., 2020). A parallel and instructive line of work has subjected widely-held “folk momentum” beliefs to quasi-experimental scrutiny, often deflating them: the supposed advantage of scoring just before half-time is contested (Gauriot and Page, 2018; Meier et al., 2020a; Baert and Amez, 2018), the notion that a two-goal lead is uniquely dangerous does not survive careful analysis (Kent and Brandes, 2025), and apparent “success breeds success” effects can be largely mechanical rather than psychological (Gauriot and Page, 2019), though physical correlates of scoring and conceding are detectable in running output (Konefal et al., 2024). This skepticism has deep roots in the “hot hand” literature, which cautioned that perceived streaks are frequently indistinguishable from chance (Gilovich et al., 1985; Bar-Eli et al., 2006) before a selection-bias correction partially rehabilitated the effect (Miller and Sanjurjo, 2018); separating genuinely psychological momentum from strategic responses to the score remains the central identification challenge (Meier et al., 2020b), one that in-game win-probability models must also confront (Robborechts et al., 2021).

The design draws on the case-crossover method of Maclure (1991), in which each case serves as its own control by comparing a hazard window to referent windows drawn from the same subject, thereby eliminating all time-invariant confounding by construction. Because validity hinges on how referents are chosen, we follow the methodological cautions of Mittleman et al. (1995) and Marshall and Jackson (1993), and in particular the literature on time-trend bias: Lumley and Levy (2000) and Janes et al. (2005) show that naive referent schemes can bias estimates in the presence of exposure trends, with time-stratified selection the preferred and essentially bias-free strategy (Mittleman, 2005). The design is related to the self-controlled case series (Farrington, 1995; Whitaker et al., 2006) and to segmented and controlled interrupted time series methods (Wagner et al., 2002; Lopez Bernal et al., 2017, 2018). We frame the estimand within the potential-outcomes account of causation (Rubin, 1974; Holland, 1986; Hernán and Robins, 2020), making explicit the exchangeability, positivity, and SUTVA assumptions required to interpret a within-match break contrast causally. For estimation we exploit the equivalence between within-subject (fixed-effects) and correlated-random-effects specifications (Mundlak, 1978), with cluster-robust and generalized-estimating-equation inference respecting within-match clustering (Liang and Zeger, 1986; Cameron and Miller, 2015); where the response is a trajectory around the break rather than a scalar, functional regression provides a natural extension (Morris, 2015; Goldsmith et al., 2011).

1.2 The measured outcome and a first look at the data

Our outcome derives from Attack Momentum, a proprietary minute-resolved index published by SofaScore (SofaScore, 2026) that summarises, on a signed scale, which team is generating sustained attacking pressure. We orient it so that positive values favour the home team and negative values the away team. Because the tactical claim concerns the side that is dominant at the break, which is sometimes the home team and sometimes the away team, a raw signed outcome conflates two questions. We therefore use a sign-adjusted (dominance-oriented) outcome. For a break at match minute c in match i , let

$$o_i = \text{sign}\left(\frac{1}{5} \sum_{k=1}^5 M_i(c - k)\right) \in \{+1, -1\} \quad (1)$$

be the sign of the mean raw momentum M_i over the five minutes preceding the break. The oriented momentum is $Y_i(t) = o_i M_i(t)$, so that a positive value always denotes momentum toward the pre-break dominant side (Figure 1).

The Attack Momentum series is indexed by nominal match minute, and the match clock runs through a hydration break, with the lost time added on later. Consequently the series carries

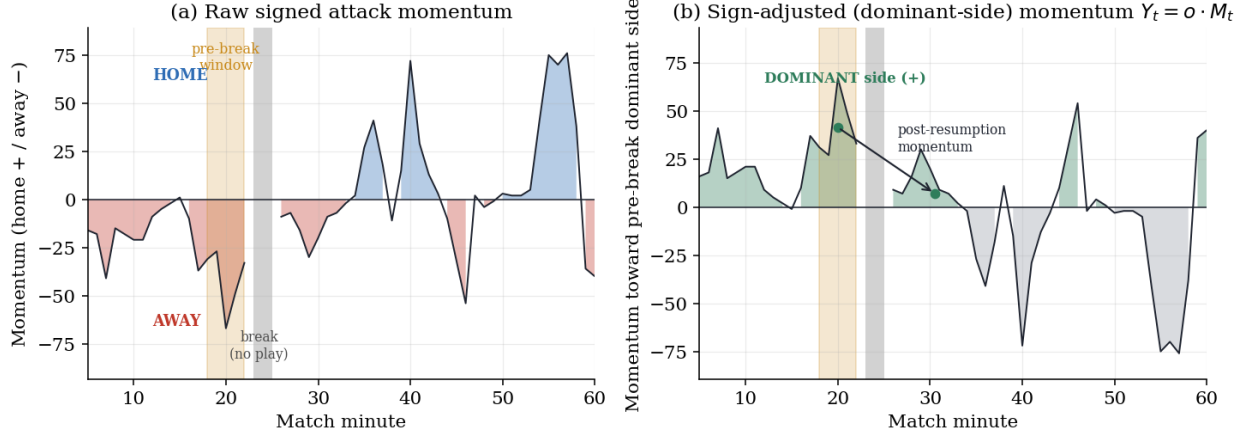


Figure 1: Constructing the sign-adjusted outcome, with the break shown as a stoppage. (a) Raw signed attack momentum for a representative match; the away side dominates the pre-break window (gold). The hatched grey band is the approximately three-minute break, during which play is stopped and the momentum series carries no informative value, and is therefore blanked. (b) After multiplication by the sign o of the pre-break mean, momentum is expressed toward the pre-break dominant side (positive by construction); the effect of interest is the change from the pre-break level to the post-resumption level (arrow).

ordinary-looking values during the break, even though play is stopped; these are algorithmic decay artefacts rather than real play, and treating them as post-break momentum would contaminate the outcome. We therefore exclude a uniform three-minute in-play window at each break start (minutes $c, c+1, c+2$) and measure the post-break outcome only from resumption, over $[c+3, c+12]$; the pre-break window is $[c-5, c-1]$. Observed break widths are tightly concentrated (96% are two or three minutes; Section 2), which justifies the uniform exclusion. The same three-minute pseudo-break exclusion is applied to every control anchor, so that the treated-versus-control contrast preserves an identical temporal structure.

A naive reading of Figure 1(b) suggests that the break was effective, since the dominant side is weaker afterward. Momentum is mean-reverting, however, and this before-and-after contrast is precisely what a sound design must resist: Section 3 shows that the dominant side’s momentum reverts through the stoppage as if play had continued, decaying at essentially the same rate whether or not a break occurs, so the break neither creates nor prevents the fade. What it removes instead are the interpolated minutes of play in which the dominant side would have pressed its advantage, a reading we develop in Section 6 alongside an event-based check that finds no effect on chance creation.

1.3 Two features of the policy, and contributions

Two structural facts shape the analysis. First, breaks occur only in two narrow windows: a first-half break with median minute 23 and a second-half break with median 68 (Figure 2). As Section 4 makes precise, the design identifies the effect of a break at the minute at which it was administered, and cannot, in the absence of an untestable homogeneity assumption, speak to a break at some other minute. Second, the policy breaks in every match, so that, unlike historical tournaments in which a stoppage of this kind occurred only in hot conditions, heat and the administration of a break are not separable in any across-match comparison. This entanglement motivates a purely within-match design, in which such match-level factors are differenced out rather than modelled.

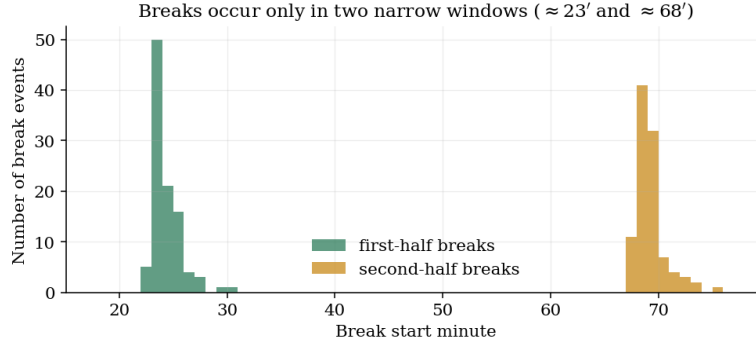


Figure 2: Breaks occur only in two narrow windows ($\approx 23'$, $\approx 68'$). The estimand is local to these minutes.

We make two contributions. First, we provide a design-based answer to a widely-asserted tactical claim: the average break effect on the dominant side’s momentum is small, but when read by effect size and compatibility interval rather than by statistical significance, the data remain compatible with competitively-relevant, state-dependent effects, concentrated on leading teams, that we do not dismiss. Second, we characterise what a case-crossover design can identify when treatment recurs at fixed times. The break main effect is recoverable only by extrapolating a within-match time trend into a window that contains no untreated observations, and is correspondingly fragile, whereas break-by-covariate interactions are identified from cross-sectional variation among the treated and are stable.

The remainder of the paper is organised as follows. Section 2 describes the data and the cleaning steps. Section 3 sets out the two candidate designs for the break-window counterfactual. Section 4 develops the within-match case-crossover, its identification and its estimates, and Section 5 the external-control robustness with its different estimand. Section 6 interprets the results and Section 7 discusses their broader implications.

2 Data

2.1 Data collection

For every 2026 World Cup match with a published Attack Momentum graph we retrieved the minute-resolved series from SofaScore (SofaScore, 2026), together with goals and their minutes. The series is a signed index (home positive, away negative) posted at nominal-minute resolution: a regulation match yields approximately 92 points (the 90 nominal minutes plus a short trailing stub), with half time at the 45th, and an extra-time match approximately 124. The nominal indexing implies that the series does not carry a separate slot for every second of first-half stoppage time, which the clock accrues during a break and the referee adds after the 45th minute. Because breaks occur at nominal minutes ≈ 23 and ≈ 68 , well inside regulation play, this feature does not affect their placement.

Exact break minutes were read from live match commentary rather than assumed at a fixed minute. For each match we located the commentary markers that open and close the hydration break, yielding a start minute for each break (and, for group-stage matches, an end minute). The resulting break minutes vary across matches around the nominal 22’ and 67’ targets (Figure 2). Each event further carries the current score margin and the pre-break momentum level and slope, all computed from the series, together with match-dated international Elo for both teams, a WBGT

heat index derived from venue coordinates and kickoff time via a reanalysis weather service, and the local kickoff hour. Elo and WBGT are match-level constants; they play no confounding role in the within-match design (Section 4) but serve as effect modifiers and for description.

We verified the minute alignment of the series in three ways. First, the hydration break leaves a visible signature: momentum decays monotonically from the annotated break-start minute over the following few minutes, consistent with a lull in which no play occurs, and this decay begins at the commentary-derived break minute, confirming that break timing and the momentum index share the same axis. Second, we cross-validated against goals: in about 80% of first-half goals and 65% of second-half goals the momentum in the minute of the goal points toward the scoring side, with the remaining mismatches being against-the-run-of-play goals (a counter-attack converted while the conceding side held the momentum) rather than an indexing offset. The somewhat lower second-half agreement is consistent with the greater accumulation of stoppage time later in the half, and does not affect the near-23' and near-68' break windows on which the analysis relies. Third, we gated on series length, as described below.

2.2 Cleaning

Four cleaning steps were applied. First, two 2022-reference venues had transposed coordinates that returned implausible temperatures; these were corrected before WBGT computation. Second, climate-controlled or domed stadia (Atlanta, Houston, and Arlington, together with Las Vegas and Glendale) were assigned an indoor WBGT of 20.7°C keyed on city name, since outdoor reanalysis does not reflect the indoor environment. Third, the three in-play minutes at each break were blanked and the post-break outcome measured from resumption, as described in Section 4. Fourth, because the analysis relies on nominal-minute indexing, we excluded any match whose momentum series length is inconsistent with that indexing: one match (Mexico versus South Africa) returned an empty series, and two group-stage matches (Argentina versus Algeria, 136 points, and Austria versus Jordan, 132 points) returned over-long series carrying interpolation and padding artefacts that would displace the minute axis and the break location. These three matches were dropped.

One further match, France versus Iraq (group stage, match identifier 15186769), had only a first-half break recorded in commentary, its second half having featured no hydration break. Rather than retain a match that contributes a single break, we drop it. The result is a balanced panel of 99 matches with exactly two breaks each, that is, 198 break events, in which every retained match contributes one first-half and one second-half break.

2.3 Descriptive statistics

Table 1 summarises the analytic sample: 99 matches (67 group, 32 knockout), 198 break events, and 3,139 non-break control anchors. Break start minutes are tightly clustered (first half median 23, IQR 23–24; second half median 68, IQR 68–69), and observed break widths are almost all two to three minutes. Conditions span a wide range (WBGT 17.8–34.7°C), and at the break the pre-break dominant side is most often level (98 of 198) but frequently leading or trailing, which supplies the cross-sectional scoreline variation that identifies the interaction terms.

3 Two designs for the break-window counterfactual

Identifying the break effect requires an assumption about what momentum would have done during the break minutes, which are excluded from the outcome. Two natural counterfactuals are available,

Table 1: Descriptive statistics for the analytic sample (99 matches, 198 break events).

Quantity	Value
Matches (group / knockout)	99 (67 / 32)
Break events (one per half)	198
Non-break control anchors	3,139
First-half break start minute	median 23 (IQR 23–24)
Second-half break start minute	median 68 (IQR 68–69)
Break width	median 3 min; 96% are 2–3 min
WBGT heat index (AC venues corrected)	mean 24.2°C (SD 4.3), range 17.8–34.7
Air-conditioned-venue matches	24
Local kickoff hour	median 17:00 (range 12–22)
Pre-break dominant-side lead state	behind 41; level 98; ahead by 1 40; ahead by ≥ 2 19
Oriented Elo gap of dominant side	mean +111 (SD 227)
Pre-break oriented momentum level	mean +20.4

and we report the analysis under both (Figure 3). They differ only in how a break event’s post-break window is matched to a control’s post-anchor window.

Under the clock-aligned counterfactual, momentum would have continued to evolve on the game clock during the stoppage, so a break event and a control are compared at the same clock offset from their anchor: for a break at start minute s the outcome is the oriented mean momentum over $[s+3, s+12]$, and for a control at minute c over $[c+3, c+12]$, both beginning three minutes after the anchor. Under the play-aligned counterfactual, the break is dead time during which momentum is suspended, so the resume minute is the first minute of play and is matched to a control’s first minute of play: the break event’s outcome is the oriented mean over the ten play-minutes from its resume minute e (the break end, taken per match), $[e, e+9]$, and the control’s is over its immediate continuation $[c+1, c+10]$. The break event’s window is identical under both counterfactuals; only the control window changes, advanced by the break duration under the play-aligned assumption. Covariates and the game-time trend are evaluated at the anchor minute in both.

The two designs give different pictures of the same data, shown side by side in Figure 4. Under the clock-aligned design (panel a), a break event and a control are compared at the same clock offset, and the two decay in near-lockstep from about +20 before the break to about +8 after it: the dominant side’s momentum reverts at the same rate whether or not a break occurs, so the break neither creates nor prevents the fade, which unfolds through the stoppage as if play had continued. Under the play-aligned design (panel b), the break is removed and the resume minute is treated as the first minute of play, so a break event is compared with a continuous-play control at the same amount of play; here the break events sit below the controls for the first few play-minutes before converging. That transient gap, present only under the play-aligned design, is confined to the window in which a decaying momentum index under-reads after a stoppage, and Section 6 shows with an event-based outcome that it reflects the index rather than the team.

4 The within-match case-crossover

4.1 The case-crossover target

Index a candidate intervention point by (i, c) , denoting match i and minute c . Let $D_{ic} \in \{0, 1\}$ denote a break at (i, c) , and let $Y_{ic}(1), Y_{ic}(0)$ be the potential sign-adjusted post-resumption momenta. In 2026 a break is administered at the two minutes $c \in \{c_{i1}, c_{i2}\}$ near 23’ and 68’, and

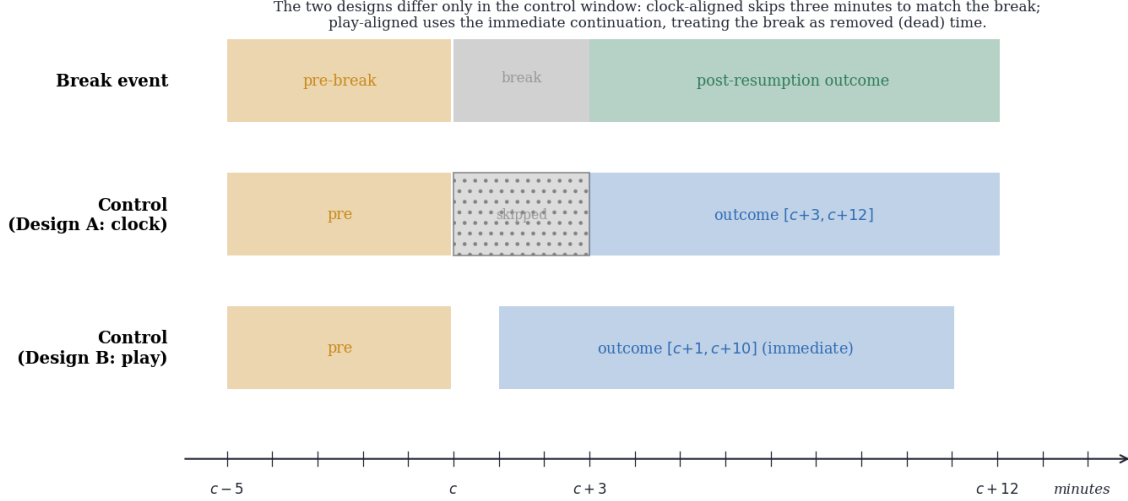


Figure 3: The two counterfactuals for the break window. A break event’s post-resumption outcome window (green) is the same under both. Under the clock-aligned design a control skips three minutes to match the break and its outcome begins at $c+3$; under the play-aligned design the control uses its immediate continuation from $c+1$, treating the break as removed dead time. The two designs differ only in the control window.

all other within-match minutes are untreated. The target is the case-crossover average treatment effect on the treated, local to the observed break minutes,

$$\tau = \mathbb{E}[Y_{ic}(1) - Y_{ic}(0) \mid D_{ic} = 1]. \quad (2)$$

The case-crossover design (Maclure, 1991) takes each treated unit as its own control by comparison to referent untreated occasions from the same subject, namely non-break minutes in the same match, so that all stable subject-level characteristics cancel; the choice of referent minutes is the design’s key degree of freedom (Janes et al., 2005).

4.2 Identifying assumptions

Write the within-match adjustment set as $X_{ic} = \{g(c), \text{half}, L_{ic}, S_{ic}\}$, where $g(c)$ is a smooth game-time trend, L_{ic} the pre-break momentum level and slope, and S_{ic} the scoreline state. Identification of (2) rests on four assumptions. The first (A1) is conditional exchangeability given X_{ic} and a match effect α_i : within a match, whether minute c is a break minute is governed by the clock and the referee’s protocol rather than by the evolving state of play once game time and momentum state are conditioned on; as a specificity check on this assumption, a placebo break placed at 38’/82’ returns no effect (Appendix Appendix C), indicating that the estimator does not manufacture effects at arbitrary non-break minutes. The second (A2) is no anticipation, in that pre-break play is not a response to the upcoming break. The third (A3) is SUTVA, that is, no cross-match interference and a single version of the treatment. The fourth (A4) is within-match positivity, in that referent minutes with comparable X_{ic} exist; this holds in the scoreline and momentum-state dimensions but not in game time, since no untreated minute exists at the break location (Section 4.4). Under (A1)–(A4), the adjusted within-match contrast of post-resumption momentum between break and referent minutes identifies τ (Rubin, 1974; Hernán and Robins, 2020).

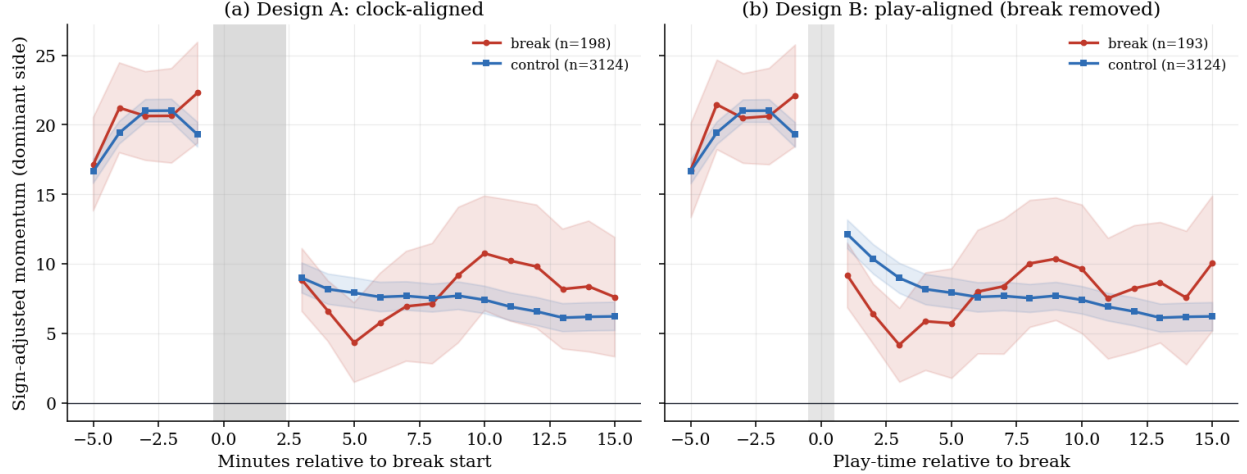


Figure 4: What the two designs show. Event-time average of sign-adjusted momentum for break events (red) and controls (blue), oriented toward the pre-break dominant side. (a) Clock-aligned: both series are aligned on the game clock and the in-break minutes are blanked; the two decay alike, so the break neither creates nor prevents the fade. (b) Play-aligned: the break is removed and the resume minute is play +1; the break events dip below the continuous-play controls for the first few play-minutes, then converge. Bands are ± 1.96 pooled standard errors.

4.3 Match-level covariates are absorbed by design

A decisive feature of the self-matched design is that every characteristic fixed at the match level is differenced out mechanically, whether or not we measure it. The two teams, their Elo ratings and Elo gap, the WBGT, the kickoff hour, the stadium, and the tournament era are all constant across the break and referent minutes of a match; they are collinear with α_i and cannot confound the within-match contrast. In particular, Elo cannot enter as a confounding covariate: as a regressor alongside α_i it is perfectly collinear and is dropped. Elo re-enters legitimately only as an effect modifier, $D_{ic} \times \text{Elo gap}_i$, identified from cross-sectional Elo variation across matches, and we include it in Section 4.6.

4.4 The estimand is local to the break minute

Because breaks occur only near 23' and 68' and we exclude the in-play window around each, there is no untreated observation at the break location. The no-break momentum against which the main effect is compared is therefore never observed; it is extrapolated from referent minutes elsewhere in the match via $g(c)$ and the state controls. Two consequences follow.

The first is that the break main effect β is a time-trend extrapolation and is therefore fragile. Since β leans on the shape of $g(c)$ carried into the unobserved window, it moves when g is respecified or when the referent minutes anchoring it are relocated (Table 5, in Appendix B). Polynomial trends leave β near zero, but saturated per-minute dummies, which absorb all game-time variation and leave nothing to anchor the extrapolation, send β to -1.7 with an interval of roughly ± 7 ; relocating the referent band moves β from $+1.1$ to -5.1 . The main effect is thus identified only under a smoothness assumption on $g(c)$.

The second is that the interaction γ , by contrast, is contrast-identified and stable. The break-by-scoreline interaction is the difference between how the outcome depends on the scoreline at break minutes and how it depends on the scoreline at referent minutes, a contrast formed within

each location from cross-sectional variation in the scoreline among the break events. It does not rely on the time-trend extrapolation, and accordingly it sits near -0.2 across trend specifications and, apart from the thin distal “far” referent band, is far more stable across referent windows than β (Table 5). In this sense the design can speak to how the break interacts with match state even though it only weakly pins down the average level effect, and it cannot speak at all to a break at an arbitrary minute without assuming homogeneity over game time, an assumption that is untestable here and substantively implausible.

4.5 The model

Our primary estimator is a within-match fixed-effects analysis of covariance with match-clustered standard errors, fit on break and referent minutes jointly:

$$Y_{ic} = \alpha_i + g(c) + \phi \text{half}_c + \lambda L_{ic}^{\text{lev}} + \psi L_{ic}^{\text{slp}} + \delta S_{ic} + \beta D_{ic} + \gamma (D_{ic} \times S_{ic}) + \boldsymbol{\eta}^\top (D_{ic} \times \mathbf{Z}_i) + \varepsilon_{ic}, \quad (3)$$

where α_i is a match fixed effect; $g(c) = \theta_1 c + \theta_2 c^2$ (quadratic by default); L_{ic}^{lev} and L_{ic}^{slp} are the oriented pre-break level and slope; S_{ic} is the oriented score margin; D_{ic} is the break indicator; and $\mathbf{Z}_i$ collects match-level modifiers (heat and Elo gap) entered only in interaction. Standard errors are clustered on match i (Liang and Zeger, 1986; Cameron and Miller, 2015). Because the outcome is sign-adjusted, most between-match variation is removed and the residual intraclass correlation is small and boundary-unstable; we therefore treat the fixed-effects estimator as primary and report a random-intercept (Mundlak-style) mixed model (Mundlak, 1978) as a corroborating check. The match fixed effect α_i enforces the case-crossover logic exactly, and $g(c)$ makes break and referent minutes comparable with respect to the natural evolution of momentum over the match (Figure 4); under (A1)–(A4), β together with the interaction terms is a plug-in estimate of τ , subject to the locality caveat of Section 4.4.

4.6 Estimates: a small average effect under both counterfactuals

We report and interpret effect sizes and their compatibility (confidence) intervals rather than declaring results significant or null, following widely-endorsed guidance that a large p -value is not evidence of no effect and that an interval should be read for the range of effects with which it is compatible (Wasserstein and Lazar, 2016; Greenland et al., 2016; Amrhein et al., 2019). This stance matters here for a substantive reason: the policy is intended to be competitively neutral, so the quantity of interest is not only the average effect but whether the break moves momentum in any match state by an amount that would be material to the contest. For orientation, the pre-break dominant side sits at about $+20$ momentum points and regresses naturally to about $+8$ over the outcome window (Figure 4); an effect of a few points is therefore a non-negligible fraction of the match’s natural swing rather than statistical noise.

Table 2 reports the full nested models under the clock-aligned counterfactual, and Table 4 (in Appendix B) the play-aligned counterfactual. The average break effect is small under both: the baseline break coefficient is $+0.26$ (cluster-robust standard error 1.41) under the clock-aligned design and -0.63 (standard error 1.53) under the play-aligned design. Both are centred near zero with intervals wide relative to the scale. With only 99 matches, however, the design has limited power to detect a small average effect, so these wide intervals are as much a statement about power as about the absence of an effect; the appropriate reading is “no evidence of a large average effect” rather than “evidence of no effect,” and the informative content is the direction of the estimates rather than their statistical significance. The two designs differ only in the sign of a small, imprecisely-estimated main effect and agree almost exactly on the covariate coefficients and,

Table 2: Nested within-match fixed-effects models, clock-aligned counterfactual (outcome over $[c+3, c+12]$ for both break events and controls). Coefficients with cluster-robust standard errors in parentheses. The break $\times$ lead term is a single continuous slope on the oriented goal margin; the break $\times$ Elo term is per 100 Elo points. Match fixed effects included but not shown.

Term	Baseline	+ Break $\times$ lead	+ Break $\times$ Elo	Full
Minute	−0.09 (0.16)	−0.09 (0.16)	−0.09 (0.16)	−0.09 (0.16)
Minute ² ($\times 10^3$)	+0.85 (1.27)	+0.83 (1.27)	+0.84 (1.27)	+0.84 (1.27)
Second half	+2.41 (4.21)	+2.37 (4.21)	+2.39 (4.20)	+2.40 (4.20)
Pre-break level	+0.24 (0.04)	+0.24 (0.04)	+0.23 (0.04)	+0.23 (0.04)
Pre-break slope	+0.06 (0.03)	+0.06 (0.03)	+0.06 (0.03)	+0.06 (0.03)
Score margin	−2.31 (1.20)	−2.23 (1.27)	−2.24 (1.26)	−2.23 (1.26)
Break	+0.26 (1.41)	+0.38 (1.49)	+0.36 (1.49)	+0.35 (1.49)
Break $\times$ lead	−	−0.90 (1.39)	−2.23 (1.36)	−2.19 (1.37)
Break $\times$ Elo/100	−	−	+1.59 (0.80)	+1.58 (0.81)
Break $\times$ heat/ $^{\circ}$ C	−	−	−	−0.21 (0.31)
Match fixed effects	Yes	Yes	Yes	Yes
Observations	3337	3337	3337	3337
Matches	99	99	99	99

as shown below, on the interaction structure. A random-intercept mixed model returns nearly identical break-term estimates, consistent with a near-zero intraclass correlation.

4.7 Heterogeneity by lead and by heat

The average masks systematic variation with match state, which is precisely the aspect on which a competitive-neutrality concern should focus, and the interaction structure is nearly identical across the two counterfactuals. The lead interaction is a single continuous slope on the oriented goal margin rather than a set of categorical cells: the estimated break effect changes by about -0.9 momentum points per goal of lead when averaged over team strength, and by about -2.4 per goal once team strength is held fixed, in both counterfactuals and using all 198 break events (Tables 2 and 4). The two slopes differ because lead and strength are correlated, since the side that is both dominant and ahead is disproportionately the pre-match favourite, so that each modifier masks the other until both are included. The break $\times$ Elo term is positive (about $+1.6$ per 100 Elo), indicating that a stronger in-form side sustains momentum more.

Figure 5 plots the implied break effect as a function of the oriented goal margin, with a 95% compatibility band. At a level scoreline the effect is essentially zero ($+0.4$); it is mildly positive for a trailing in-form side and negative for a leading one, reaching an implied -1.4 averaged over strength, or -4.1 at average strength, at a two-goal lead. The band widens toward large margins because few break events occur there, so the effect in that region is an interpolation along the fitted slope rather than an independent estimate. At a two-goal lead the strength-adjusted interval is approximately $[-9.3, +1.1]$, which is asymmetric about zero: it extends to -9.3 on the harm side but only to $+1.1$ on the benefit side, so the data are considerably more compatible with the break eroding a leading side’s momentum than with enhancing it.

Heat is the second candidate modifier. Splitting the within-match sample at a wet-bulb globe temperature of 28°C , the break effect in cool matches is near zero under both designs ($+0.5$ clock-aligned, -0.3 play-aligned), whereas in hot matches it is negative (-0.96 , interval $[-6.6, +4.7]$, clock-aligned; -2.80 , interval $[-8.3, +2.7]$, play-aligned; $n = 42$ hot break events). The hot-match estimate leans against the dominant side, more so under the play-aligned counterfactual, but its

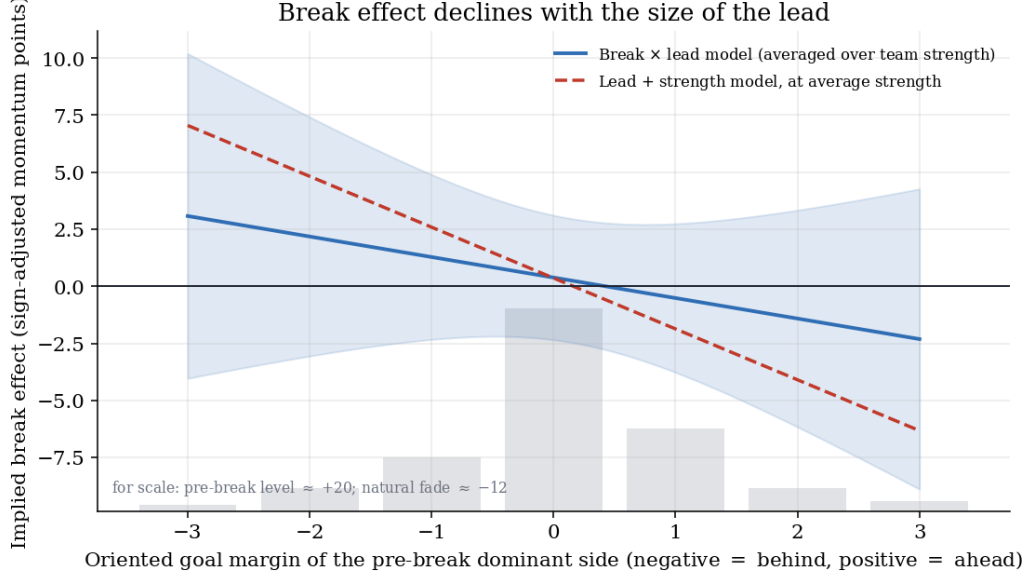


Figure 5: Implied break effect on the in-form side as a function of the oriented goal margin. The solid line is the continuous break $\times$ lead slope averaged over team strength; the dashed line holds team strength at its mean. The shaded region is the 95% compatibility band for the solid line, which widens away from a level scoreline because that is where most break events occur (grey bars, indicating data support). Negative values denote momentum moved against the pre-break dominant side.

interval spans zero. We return to this in Section 6, where the same pattern appears, less precisely, in the external design.

The state-dependent structure is exploratory: the large-lead and hot cells are thin (a lead of two or more occurs 19 times, and 42 break events are hot), the modifiers were selected after inspection, and no multiplicity adjustment is applied. It is a hypothesis for replication under the same universal-break policy rather than a confirmed effect. Robustness and falsification checks for the within-match design, comprising the stability of the interaction across trend and referent choices, a placebo break at 38'/82', and sensitivity to the width of the in-play exclusion, are reported in Appendix C and are consistent with the conclusions above.

5 Robustness: external controls

5.1 A different estimand and its data

The within-match design compares a break minute to other minutes of the same match. A complementary design compares each 2026 break event to no-break minutes drawn from other matches entirely. The estimand it targets is different: a between-match contrast that no longer differences out the teams, the venue, the heat, or the era, and therefore requires those factors to be adjusted for rather than eliminated by design. Writing the external adjustment set as $X_{ic}^{\text{ext}} = \{Z_i, L_{ic}, S_{ic}, \text{half}\}$, where the match-level modifiers Z_i (Elo, WBGT, and local kickoff hour) now enter directly because they are no longer differenced away, the design targets the same treated contrast as (2),

$$\tau_{\text{ext}} = \mathbb{E}[Y_{ic}(1) - Y_{ic}(0) \mid D_{ic} = 1], \quad (4)$$

but must recover the counterfactual $Y_{ic}(0)$ from no-break minutes in other matches rather than from the same match. Under between-match conditional ignorability, $Y_{ic}(0) \perp D_{ic} \mid X_{ic}^{\text{ext}}$, and

positivity, $0 < \Pr(D_{ic} = 1 \mid X_{ic}^{\text{ext}}) < 1$,

$$\tau_{\text{ext}} = \mathbb{E}[\mathbb{E}[Y_{ic} \mid D_{ic} = 1, X_{ic}^{\text{ext}}] - \mathbb{E}[Y_{ic} \mid D_{ic} = 0, X_{ic}^{\text{ext}}] \mid D_{ic} = 1], \quad (5)$$

which the tight-caliper matching of Section 5.3 estimates on the region of common support. The positivity condition in (5) is precisely what fails for the extreme-heat, early-afternoon events (Section 5.2). We assemble no-break control minutes from historical tournaments that did not administer universal breaks, namely Copa América (2019, 2021, 2024), the UEFA European Championship (2020, 2024), the CONCACAF Gold Cup (2019, 2021, 2023, 2025), and the 2018 and 2022 World Cups, retaining minutes in the same windows in which 2026 breaks occur and carrying the same covariates: match-dated Elo, WBGT (with the same air-conditioning correction), local kickoff hour, score margin, and pre-break momentum level and slope. All quantities are oriented toward the pre-break dominant side, exactly as in the within-match design.

5.2 The positivity limitation

This design has an identification limit that the within-match design does not. Because the 2026 policy breaks in every match whereas historical stoppages of this kind occurred only in hot conditions, the extreme-heat and early-afternoon games the policy most plausibly targets have essentially no historical no-break counterfactual. Figure 6 in Appendix A shows the overlap: the historical control pool has no match above a WBGT of 32°C, above which a dozen 2026 break events lie, and 28% of break events kick off at or before 14:00 against 2% of controls. The extreme-condition regime is therefore a positivity void, and any external-control estimate there rests on extrapolation. This is the central reason we treat the external analysis as a robustness check rather than a primary result, and it motivates trimming to common support.

5.3 Effect estimates under tight-caliper matching

We estimate the effect by tight-caliper propensity-score matching (Rosenbaum and Rubin, 1983; Stuart, 2010), which trims to common support rather than reweighting into the void: each break event is matched to its nearest control within a caliper of 0.2 standard deviations on the estimated propensity logit, following the caliper width recommended by Austin (2011), and events without a match inside the caliper are dropped. Only three break events are dropped, all early-afternoon games, consistent with the positivity diagnosis. On the matched support the average effect is small and not distinguishable from zero under both counterfactuals, and, as in the within-match design, more negative under the play-aligned assumption: the matched effect is -1.57 (interval $[-5.00, +1.87]$) clock-aligned and -2.48 (interval $[-5.97, +1.01]$) play-aligned. The direction agrees with the within-match crossover under the play-aligned assumption, where the effect is also negative, though every interval crosses zero.

5.4 Conditional effects

The external design is under-powered for the lead heterogeneity: balancing thin lead cells against historical controls with limited overlap yields wide and non-monotonic estimates (behind $+1.4$, level -0.1 , ahead by one $+4.6$, ahead by two or more -2.2 , all with intervals several points wide), so the within-match break by lead slope of Section 4.7 remains the better-identified estimate of that heterogeneity. The heat contrast is the one place the external design adds information, and it points the same way as the within-match split: under the play-aligned counterfactual the effect in cool matches is $+0.43$ (interval $[-2.4, +3.0]$) and in hot matches -4.60 (interval $[-9.6, +7.6]$);

these external-design intervals are nonparametric bootstrap percentile intervals and need not be symmetric about the point estimate. The hot-match point estimate is sizable and negative, but its interval is enormous, which is the positivity limitation made concrete: the hot-condition effect is suggestive but not identified from historical data.

6 Interpretation of the results

6.1 Momentum reverts through the break as if play had continued

The most instructive single result is Figure 4. Because the clock-aligned and play-aligned designs differ only in the control window, the clock-aligned comparison implicitly asks whether momentum freezes during the stoppage, and it does not: the oriented momentum of a break event falls during the break minutes at essentially the same rate as a continuous-play control, so the series behaves as though play had gone on while play was suspended for hydration. The substantive implication is at the resumption. The dominant side re-enters the match having missed the interpolated minutes of play during which, in a continuous game, it would have continued to press its advantage, and that stretch is unobservable by construction because no football is played during it. Whether the lost continuation carries any competitive cost cannot be settled by the momentum index alone, which merely decays toward zero absent new events.

An event-based outcome speaks to the question. Net expected goals (net xG) accrues only on shots and is a true zero during a stoppage rather than a decaying value, so it is free of the rebuild lag that produces the transient dip in Figure 4. Measured under both counterfactuals with the same per-match resume minutes, the break effect on net xG is essentially zero under both counterfactuals (-0.001 , interval $[-0.051, +0.049]$, clock-aligned; $+0.011$, interval $[-0.051, +0.073]$, play-aligned). The dominant side creates the same quality of chances after the break as a continuous-play comparison, whether aligned on clock-time or on play-time. The two readings should be held together: the momentum series behaves as though the break excised a stretch of the dominant side’s pressure, yet on chance creation that removal leaves no measurable trace. We therefore read the break as neither creating nor preventing the fade, while noting that the interpolated minutes it removes, though without a detectable effect on net xG, are precisely the ones we cannot observe.

6.2 Reading the effect sizes across scenarios

Table 3 collects the break main effect across designs and counterfactuals. Every interval crosses zero, but they are not all equally informative. The within-match clock-aligned estimate is small and near zero, with an interval of about ± 3 momentum points; on a scale whose natural swing over the window is about 12 points, an average effect larger than a few points is not well supported. We are cautious, however, not to read this as positive evidence of a null: with only 99 matches the design has limited power to detect a small average effect, so an interval that includes zero reflects that limited power as much as an absent effect. The play-aligned estimates lean negative in every design, and the tight-caliper matched estimate is the most negative (-2.5); none is individually distinguishable from zero, but the consistent negative sign under the play-aligned assumption is the more informative feature.

The state-dependent estimates require the opposite reading. The break by lead slope and the hot-match contrast have wide, and in the leading-side case asymmetric, intervals, so significance testing would discard them; yet their effect sizes carry a story a competitively-neutral policy should not ignore. A comfortably-leading side of average strength has an implied effect near -4 momentum points with an interval reaching -9 and barely crossing zero, and hot matches lean negative in

Table 3: Break main effect on the dominant side’s momentum across designs and counterfactuals (coefficient with 95% interval). All intervals include zero; the play-aligned column leans negative.

Design	Clock-aligned	Play-aligned
Within-match case-crossover	+0.26 [−2.50, +3.02]	−0.63 [−3.63, +2.37]
External, tight-caliper matched	−1.57 [−5.00, +1.87]	−2.48 [−5.97, +1.01]

both the within-match (−2.8 play-aligned) and external (−4.6) designs. These are not findings in the confirmatory sense, but their consistent sign and appreciable magnitude mark exactly the conditions, a large lead or extreme heat, in which a break could plausibly move a match, and in which the extreme-heat case is furthermore unidentifiable from historical data.

6.3 Bringing the two designs together

The within-match and external designs agree on the substance. The average break effect is small and not distinguishable from zero under both, and the play-aligned assumption produces a mild negative lean in both. The lead heterogeneity is cleanly estimated only within match, where the effect declines with the size of the lead; the external design is too thin here to add more than imprecise corroboration. The heat heterogeneity is where the two designs reinforce each other in direction, both placing the hot-match effect against the dominant side, while differing in precision, and it is also where the external design’s positivity void forecloses identification. The coherent picture is of a policy with no detectable average effect and no detectable effect on chance creation, but with a state-dependent tendency, concentrated on leading teams and hot matches, that the data cannot rule out and, in the extreme-heat case, cannot resolve.

7 Discussion

The strong form of the popular claim, that in-match hydration breaks reliably blunt the momentum of whichever side is dominant, finds little support in these data: much of what appears to be a break-induced fade is momentum reverting through the stoppage as if play had continued, equally at non-break minutes, so the break neither creates nor prevents the fade but forgoes the interpolated minutes of pressure the dominant side would otherwise have generated, a removal that leaves no trace in net xG. The average break effect is small once the in-play stoppage is excluded and match-level factors are differenced out by the case-crossover design. We are careful, however, not to read this as evidence that the break has no effect. With only 99 matches, and two breaks each, this sample does not have the power to reject a small average effect, so our failure to detect one reflects limited power as much as an absent effect; a wide compatibility interval that includes zero cannot establish that the effect is zero, only that a large effect is not well supported. Read by effect size and compatibility interval, the data continue to admit momentum shifts of a few points and, more importantly for a policy intended to be competitively neutral, the effect is not uniform: for a comfortably-leading, average-strength side the implied effect is about −4 momentum points, with an interval that reaches −9 and only marginally crosses zero, so that the data are considerably more compatible with the break eroding a leading side’s momentum than with its enhancing it. This is a state-dependent, competitively-relevant effect that cannot be ruled out, and because the policy aims for no match to be affected, it warrants monitoring and replication rather than dismissal. Methodologically, when an intervention recurs at fixed times its average level effect is only weakly identified, resting on a trend extrapolation into a window that contains no untreated data, whereas

its interactions with match state are cleanly identified from cross-sectional variation. The state-dependent signal, rather than the average, is therefore where both the statistical information and the policy relevance reside. These conclusions hold under both counterfactuals for the break window, the clock-aligned and the play-aligned, and on an event-based outcome, xG, that is invariant to that choice (Section 6); the external-control design agrees on the small, indistinguishable-from-zero average but cannot identify the extreme-heat regime the policy most plausibly targets.

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Appendix A Positivity of the external-control design

Figure 6 displays the overlap between the 2026 break events and the historical no-break control minutes in the heat-by-kickoff plane. The historical pool contains no match above a WBGT of 32°C, above which twelve break events lie, and only 2% of control minutes kick off at or before 14:00 against 28% of break events. In precisely the extreme-heat, early-afternoon regime that the mandatory-break policy most plausibly targets, there is no historical no-break counterfactual, so the external-control estimate there is an extrapolation rather than an identified contrast.

Appendix B Supplementary tables

This appendix collects two tables referenced from the main text: the full nested within-match models under the play-aligned counterfactual (Table 4), the companion to the clock-aligned Table 2; and the referent- and time-trend-sensitivity analysis (Table 5) supporting the identification argument of Section 4.4.

Appendix C Robustness and falsification checks for the within-match design

The interaction estimates are stable across game-time trend specifications and, apart from the thin distal “far” referent band, far more stable across referent windows than the main effect (Table 5), consistent with contrast identification, whereas the break main effect is not, consistent with extrapolation. A placebo break at 38’/82’ returns no effect, indicating that the estimator does not manufacture effects at non-break minutes. The result is also insensitive to the width of the in-play

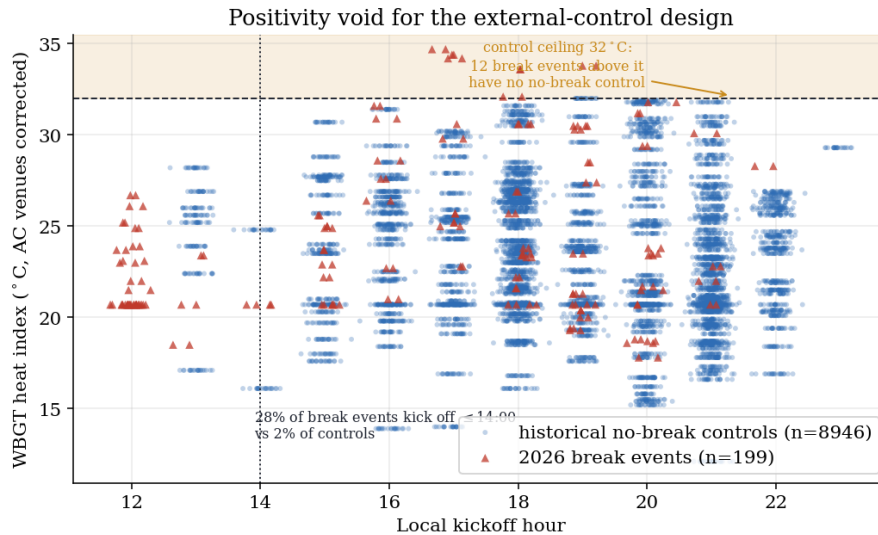


Figure 6: Positivity void for the external-control design. 2026 break events (red) and historical no-break control minutes (blue) by WBGT and local kickoff hour. The control pool has no match above the dashed 32°C ceiling and few before 14:00, so the hottest and earliest break events have no comparable control.

exclusion: widening it from three to four or five minutes leaves the average break effect small and centred near zero (+0.26, +0.64, +1.50; all intervals wide and spanning zero) and the interaction structure unchanged.

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Table 4: Nested within-match fixed-effects models, play-aligned counterfactual (outcome over ten play-minutes from resumption for break events and over the immediate ten-minute continuation for controls, using each match’s actual break start and resume minutes). Cluster-robust standard errors in parentheses. Compare with Table 2.

Term	Baseline	+ Break×lead	+ Break×Elo	Full
Minute	+0.03 (0.16)	+0.03 (0.16)	+0.03 (0.16)	+0.03 (0.16)
Minute ² ($\times 10^3$)	−0.60 (1.18)	−0.61 (1.18)	−0.61 (1.17)	−0.62 (1.18)
Second half	+2.31 (4.05)	+2.26 (4.05)	+2.28 (4.03)	+2.30 (4.03)
Pre-break level	+0.23 (0.04)	+0.23 (0.04)	+0.22 (0.04)	+0.22 (0.04)
Pre-break slope	+0.16 (0.04)	+0.16 (0.04)	+0.16 (0.04)	+0.16 (0.04)
Score margin	−1.98 (1.09)	−1.89 (1.14)	−1.89 (1.14)	−1.89 (1.14)
Break	−0.63 (1.53)	−0.49 (1.59)	−0.51 (1.61)	−0.53 (1.60)
Break × lead	—	−1.04 (1.36)	−2.43 (1.31)	−2.37 (1.32)
Break × Elo/100	—	—	+1.66 (0.82)	+1.66 (0.83)
Break × heat/°C	—	—	—	−0.30 (0.31)
Match fixed effects	Yes	Yes	Yes	Yes
Observations	3355	3355	3355	3355
Matches	99	99	99	99

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Table 5: The main effect is an extrapolation; the interaction is a contrast. Gap-corrected sign-adjusted momentum over $[c+3, c+12]$. Top: varying the game-time trend $g(c)$ with all referent minutes. Bottom: fixing the break rows and relocating the referent band. The main effect β swings; the break $\times$ lead interaction γ is comparatively stable (the far band is a thin, distal slice).

Specification	Break main effect β [95% CI]	Break $\times$ lead γ [95% CI]
<i>Vary the time trend $g(c)$; all referent minutes</i>		
linear	+0.26 [−2.69, +3.21]	−0.17 [−2.77, +2.43]
quadratic	+0.50 [−2.42, +3.42]	−0.15 [−2.76, +2.46]
cubic	+0.92 [−2.37, +4.21]	−0.18 [−2.79, +2.44]
per-minute dummies	−1.70 [−9.01, +5.61]	−0.12 [−2.65, +2.41]
<i>Fix break rows; relocate referent band (H1 H2)</i>		
adjacent 15–32 60–77	−0.31 [−4.52, +3.90]	+0.55 [−1.98, +3.08]
mid 11–40 56–85	+1.07 [−2.68, +4.82]	+0.48 [−2.35, +3.30]
far 6–15 73–78	−5.14 [−13.46, +3.19]	−2.34 [−8.78, +4.10]
all eligible	+0.50 [−2.42, +3.42]	−0.15 [−2.76, +2.46]

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